

Name $\Rightarrow$ Siddhorth Shukla

Reg No $\Rightarrow$ 23FE WCSE 00812

Section $\Rightarrow$ K

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Assignment - 1

Section-A

- Q1) (a) Define random variable.
It is a variable whose possible values are outcomes of a random phenomenon. There are two main types of random variables:
(i) Discrete random variable (ii) Continuous Random Variable

- (b) If two random variable X and Y are independent, then their probability distribution can be written as the product of their individual distributions.

- (c) Write the statement of Chebyshev's Inequality.
 X is a random variable on a sample space and $E(X) = \mu$ is probability distribution. Then for any positive real number

$$P(|X - \mu| \geq r) \leq \frac{V(X)}{r^2}$$

or the prob. of finding a value of x further from the mean than r is smaller than the variance divided by r^2 .

- (d) Define distribution function.
Distribution function of a random variable X is a function that gives the prob. that the random takes a value $\leq x$.

$$F(x) = P(X \leq x)$$

- (c) Is the function defined as follows a density function?
$$f(x) = \begin{cases} e^{-x} & , x \geq 0 \\ 0 & , x < 0 \end{cases}$$

Yes, the function follows a density function because it follows non-negativity and normalization.

- (f) A discrete random variable can take countable number of values within its range.

- (g) The range of distribution function is $[0, 1]$.

- (h) If X and Y are two random variables such that their expectations exist and $P(X \leq Y) = 1$ then,

- (i) $E(X) \leq E(Y)$ (ii) $E(X) = E(Y)$
(iii) $E(Y) \geq E(X)$ (iv) None of these

- (i) The height of persons in a country is a random variable of the type -
Continuous RV

- (Q2) Two dice are rolled. Let X denote the random variable which counts the total no. of pts on ~~unpaired~~ ~~face~~ ~~upward~~ faces. Construct a table giving max. zero values of the prob. mass and draw the prob. mass. Also find the distribution function of X .

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X	Combinations	No of Outcomes	Prob $P(X=n)$
2	(1,1)	1	$1/36$
3	(1,2), (2,1)	2	$2/36 = 1/18$
4	(1,3), (2,2), (3,1)	3	$3/36 = 1/12$
5	(1,4), (2,3), (3,2), (4,1)	4	$4/36 = 1/9$
6	(1,5), (2,4), (3,3), (4,2), (5,1)	5	$5/36$
7	(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)	6	$6/36 = 1/6$
8	(2,6), (3,5), (4,4), (5,3), (6,2)	5	$5/36$
9	(3,6), (4,5), (5,4), (6,3)	4	$4/36 = 1/9$
10	(4,6), (5,5), (6,4)	3	$3/36 = 1/12$
11	(5,6), (6,5)	2	$2/36 = 1/18$
12	(6,6)	1	$1/36$

PMF Table

X	2	3	4	5	6	7	8	9	10	11	12
Prob	$1/36$	$1/18$	$1/12$	$1/9$	$5/36$	$1/6$	$5/36$	$1/9$	$1/12$	$1/18$	$1/36$

$$F(n) = P(X \leq n) = \sum_{k=2}^n P(X=k)$$

$$F(n) \text{ for } n = 2, 3, \dots, 12$$

$$F(2) = P(X=2) = 1/36$$

$$F(3) = P(X=2) + P(X=3) = 1/36 + 1/18 = 1/12$$

$$F(4) = F(3) + P(X=4) = 1/12 + 1/12 = 1/6$$

$$F(5) = F(4) + P(X=5) = 1/6 + 1/9 = 5/18$$

$$F(6) = F(5) + P(X=6) = 5/18 + 5/36 = 5/12$$

$$F(7) = F(6) + P(X=7) = 5/12 + 1/6 = 2/3$$

$$F(8) = F(7) + P(X=8) = 2/3 + 5/36 = 19/18$$

$$F(9) = F(8) + P(X=9) = 19/18 + 1/9 = 5/6$$

$$F(10) = F(9) + P(X=10) = 5/6 + 1/12 = 11/12$$

$$F(11) = F(10) + P(X=11) = 11/12 + 1/18 = 35/36$$

$$F(12) = F(11) + P(X=12) = 35/36 + 1/36 = 1$$

(9)

A diameter of electric cable, say x , is assumed to be a continuous random variable with pdf:

$$f(x) = 6x(1-x), \quad 0 \leq x \leq 1$$

Check that $f(x)$ is pdf and

(i) determine a number b such that $P(X < b) = 1/2$ ($b > 0$)

(ii) obtain exp for the cdf of x

(iii) Compute $P(1/3 \leq x \leq 2/3)$

(i) valid pdf satisfy two conditions -
non negative: $f(x) = 6x(1-x) \geq 0$ for all x in the domain $0 \leq x \leq 1$

$$\begin{aligned} \text{Normalization: } \int_0^1 f(x) dx &= \int_0^1 6x(1-x) dx \\ &= 6 \int_0^1 x dx - 6 \int_0^1 x^2 dx \\ &= 6 \times 1/2 - 6 \times 1/3 = 3 - 2 = 1 \end{aligned}$$

(ii) $P(X < b) = P(X > b)$
total prob = 1 $P(X < b) = 1/2$

$$F(b) = \int_0^b 6x(1-x) dx = 1/2$$

$$\int 6x(1-x) dx = 3x^2 - 2x^3$$

$$F(b) = (3b^2 - 2b^3) - 0 = 3b^2 - 2b^3$$

$$3b^2 - 2b^3 = 1/2 = 0$$

$$\therefore b \approx 0.647 \text{ where } P(X < b) = P(X > b)$$

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$$(iii) \quad F(n) = \int_0^n 3t + (1-t) dt$$

$$= 3n^2 - 2n^3$$

$$f(n) = \begin{cases} 0, & n < 0 \\ 3n^2 - 2n^3, & 0 \leq n \leq 1 \\ 1, & n > 1 \end{cases}$$

(iv) Conditional probability $P(n \leq 1/2 \mid 1/3 \leq n \leq 2/3)$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Let $A = n \leq 1/2$ and $B = \{1/3 \leq n \leq 2/3\}$

$$P(1/3 \leq n \leq 2/3) = F(2/3) - F(1/3)$$

$$= 2(2/3)^3 - 2(1/3)^3 - [3(2/3)^2 - 2(2/3)^3]$$

$$P(B) = 13/27$$

$$P(A \cap B) = P(1/3 \leq n \leq 1/2) = F(1/2) - F(1/3)$$

$$F(1/2) = 3(1/2)^2 - 2(1/2)^3 = 1/4$$

$$P(1/3 \leq n \leq 1/2) = 1/4 - 1/27 = 13/108$$

$$P(n \leq 1/2 \mid 1/3 \leq n \leq 2/3) = \frac{P(1/3 \leq n \leq 1/2)}{P(1/3 \leq n \leq 2/3)}$$

$$= 13/108 / 13/27 = 1/4$$

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(v) Let X be a RV with following prob. dist.

x	-3	6	9
$P(X=x)$	1/6	1/2	1/3

Find $E(X)$ and $E(X^2)$ and using laws of expectation evaluate $E(2n+1)^2$

$$E(X) = \sum x_i P(X=x_i)$$

$$E(X) = (-3) \times 1/6 + 6 \times 1/2 + 9 \times 1/3 = 11/2$$

$$E(X^2) = \sum x_i^2 P(X=x_i)$$

$$= (-3)^2 \times 1/6 + 6^2 \times 1/2 + 9^2 \times 1/3$$

$$= 3/2 + 18 + 27 = 93/2$$

$$E(2n+1)^2 = E(4n^2 + 4n + 1) = 4E(n^2) + 4E(n) + E(1)$$

$$= 4 \times \frac{93}{2} + 4 \times \frac{11}{2} + 1 = 209$$

(vi) If the Chebyshev's inequality for the random variable X is given by $P(-2 \leq X \leq 2) \geq \frac{21}{25}$, find $E(X)$ and $\text{var}(X)$

$$P(X = -4) \geq 1/4, P(X = 4) \leq \frac{1}{4}$$

$$P(X = -4) \geq 1/4, P(X = 4) \leq \frac{1}{4}$$

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$$P(-2 < X < 8) \geq 21/25$$

interval $[-2, 8]$ total length is unity

$$f(x) = \frac{-2+8}{2} = 3$$

$\therefore$ interval $[-2, 8]$ is symmetric around $E(X) = 3$

$$\Rightarrow 2k\sigma = 10 \Rightarrow k\sigma = 5$$

$$P(|X - E(X)| \leq k\sigma) \geq 1 - 4/k^2$$

$$P(-2 \leq X \leq 8) \geq 21/25$$

$$1 - 4/k^2 = 21/25$$

$$\frac{1}{k^2} = 1 - 21/25$$

$$k = 5/2$$

$$\therefore k\sigma = 5 \Rightarrow \frac{5}{2}\sigma = 5 \Rightarrow \sigma = 2$$

$$\text{Var}(X) = \sigma^2 = 2^2 = 4$$

8.6) A RV X has the following function

x	1	2	3	4	5	6	7
$P(X)$	0	k	$2k$	$2k$	$3k$	k^2	$7k^2/4$

(i) find k (ii) Evaluate $P(X \leq 6)$, $P(X \geq 6)$, and $P(2 \leq X \leq 5)$ (iii) determine the distribution function of X (iv) if $P(X \leq 6) > 1/2$, find min value of a

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(i) Total prob must sum = 1

$$0 + k + 2k + 2k + 3k + k^2 + 7k^2/4 = 1$$

$$\Rightarrow 9k + 8k^2 = 1$$

$$k = \frac{-9 \pm \sqrt{113}}{16}$$

$$k \approx 1.012, -1.23$$

$$k \text{ must be } \geq 0 \therefore k = 0.102$$

$$(ii) P(X \leq 6) = P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5) + P(X=6)$$

$$= 0 + k + 2k + 2k + 3k + k^2 + 7k^2/4$$

$$k = 0.102$$

$$P(X \leq 6) = 8 \times 0.102 = 0.816$$

$$P(X \geq 6) = P(X=6) + P(X=7)$$

$$= k^2 + 7k^2/4 = 9k^2/4$$

$$k = 0.102$$

$$= 8(0.102)^2 + 0.102 = 0.185$$

$$P(2 \leq X \leq 5) = P(X=2) + P(X=3) + P(X=4) + P(X=5)$$

$$= 0 + k + 2k + 2k + 3k = 8k$$

$$k = 0.102$$

$$= 8 \times 0.102 = 0.816$$

$$(iii) f_X(x) = P(X \leq x)$$

$$F_X(x) = 0$$

$$F_X(1 \leq x < 2) = P(X=1) = 0$$

$$F_X(2 \leq x < 3) = P(X=2) = k$$

$$F_X(3 \leq x < 4) = P(X=2) + P(X=3) = 3k$$

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$$f_x(4 \leq n \leq 5) = P(n=2) + P(n=3) + P(n=4) = 54$$

$$f_x(5 \leq n \leq 5) = P(n=2) + P(n=3) + P(n=4) + P(n=5) = 84$$

$$f_x(6 \leq n \leq 7) = P(n=2) + P(n=3) = 84 + 42$$

$$f_x(n \geq 7) = 1$$

(iv) $f_x(n) > 1/2$

$$\text{for } x=5, f_x(5) = 84 = 0.84$$

$$\text{for } n=4, f_x(n) = 54 = 0.54$$

for min $a=4$ satisfies $P(n \leq a) > 1/2$

(v) A random number x is distributed at random b/w 0 and 1 so that its prob function is $f(n) = kn^2(1-n^3)$ where k is a constant. Find the value of k . Using this value of k , find its mean and variance.

$$\rightarrow \int_0^1 kn^2(1-n^3)dn = 1$$

$$\rightarrow k \int_0^1 n^2(1-n^3)dn = 1$$

$$\rightarrow k \left[\frac{n^3}{3} - \frac{n^6}{6} \right]_0^1 = 1$$

$$\rightarrow k \left[\frac{1}{3} - \frac{1}{6} \right] = k \cdot \frac{1}{6} = 1$$

$$\Rightarrow k = 6$$

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$$E(n) = \sum n \cdot f(n) = \int_0^1 n^3(1-n^3)dn$$

$$= 6 \int_0^1 (n^3 - n^6)dn$$

$$= 6 \left[\frac{n^4}{4} - \frac{n^7}{7} \right]_0^1$$

$$= 6 \left[\frac{1}{4} - \frac{1}{7} \right] = 9/14$$

$$\text{Var}(n) = E(n^2) - [E(n)]^2$$

$$E(n^2) = \int_0^1 n^2 \cdot f(n)dn$$

$$= \int_0^1 n^4(1-n^3)dn$$

$$= 6 \int_0^1 (n^4 - n^7)dn$$

$$= 6 \left[\frac{n^5}{5} - \frac{n^8}{8} \right]_0^1 = 6 \left[\frac{1}{5} - \frac{1}{8} \right]$$

$$= 6 \left[\frac{8}{40} - \frac{5}{40} \right] = 9/20$$

$$\text{Var}(n) = \frac{9}{20} - \left(\frac{9}{14} \right)^2 = 9/196$$

(vi) If t is any real no. show that the function defined by $f(n) = e^{-t}(1-e^{-t})^{n-1}$ can represent a prob function of a random variable X assuming the values $1, 2, 3, \dots$. Find the $E(n)$ and $\text{Var}(X)$ of the distribution.

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$$p(n) = e^{-t}(1-e^{-t})^{n-1}$$

$$\rightarrow \sum_{n=1}^{\infty} p(n) = e^{-t} \sum_{n=1}^{\infty} (1-e^{-t})^{n-1}$$

$$\rightarrow \sum_{n=1}^{\infty} r^{n-1} = \frac{1}{1-r} \quad (\text{GP with } a=1, r=1-e^{-t})$$

$$\Rightarrow \sum_{n=1}^{\infty} (1-e^{-t})^{n-1} = \frac{1}{1-e^{-t}}$$

$$\Rightarrow \sum_{n=1}^{\infty} p(n) = e^{-t} \times \frac{1}{1-e^{-t}} = 1$$

$$\Rightarrow E(n) = \frac{1}{1-e^{-t}} = \frac{1}{1-(1-e^{-t})} = \frac{1}{e^{-t}} = e^t$$

$$\Rightarrow \text{Var}(n) = \frac{e}{(1-e)^2} = \frac{e^{-t}}{(e^{-t})^2} = \frac{e^{-t}}{e^{-2t}} = e^t \times e^t = e^{2t}$$

69) If x and y are independent RV with mean 2, 3 and variance 1, 2 respectively. Find the mean and variance of RV $Z = 2x - 5y$

$E(x) = 2$ $E(y) = 3$
 $\text{Var}(x) = 1$ $\text{Var}(y) = 2$

$$\text{Mean} = E(Z) = E(2x - 5y)$$

$$= 2E(x) - 5E(y)$$

$$= (2 \times 2) - (5 \times 3)$$

$$= -5$$

$$\text{Variance} = \text{Var}(Z)$$

$$= \text{Var}(2x - 5y)$$

$$= 2^2 \text{Var}(x) + 5^2 \text{Var}(y)$$

$$= 4 \times 1 + 9 \times 2$$

$$= 22$$

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610) A symmetric die is thrown 600 times. Find the lower bound for the prob of getting 80 to 120 sixes

$E(S) = np = 600 \times \frac{1}{6} = 100$

$$V(S) = npq = 600 \times \frac{1}{6} \times \frac{5}{6} = \frac{500}{6}$$

$$P(|S - E(S)| < 40) \geq 1 - \frac{1}{42}$$

$$P(15-100 < 4\sqrt{500/6} \geq 1 - 1/42)$$

$$4 = \frac{-20}{\sqrt{500/6}}$$

$$P(80 \leq S \leq 120) \geq 1 - \frac{1}{400 \times (6/500)} = 19/24$$

611) pdf of any alt = $f(n) = \frac{d(f(n))}{dn}$

$$f(n) = \begin{cases} 0 & n < 0 \\ n/2 & 0 \leq n \leq 1 \\ 1/2 & 1 \leq n \leq 2 \\ n/4 & 2 \leq n \leq 4 \\ 1 & n > 4 \end{cases}$$

$$f(n) = \frac{d[f(n)]}{dn} = \begin{cases} 0 & n < 0 \\ 1/2 & 0 \leq n < 1 \\ 0 & 1 \leq n < 2 \\ 1/4 & 2 \leq n < 4 \\ 0 & n > 4 \end{cases}$$

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$$Q12) f(n) = \begin{cases} 100/n^2, & n \geq 100 \\ 0, & \text{otherwise} \end{cases}$$

$$i) P(n \leq 150) = \int_{100}^{150} \frac{100}{n^2} dn$$

$$= \int_{100}^{150} \frac{100}{n^2} dn = -100/n$$

$$P(n \leq 150) = -\frac{100}{150} + \frac{100}{100} = 1/3$$

$$P(\text{all 3 fail}) = (1/3)^3 = 1/27$$

$$(ii) P(n > 150) = 1 - P(n \leq 150)$$

$$= 1 - 1/3 = 2/3$$

$$\text{for all 3} = (2/3)^3 = 8/27$$

$$(iii) P(150 < n < 200) = P(n > 150) - P(n > 200)$$

$$= \frac{\int_{150}^{200} 100/n^2 dn}{\int_{150}^{\infty} 100/n^2 dn} = \frac{E_{100/n^2}^{200}_{150}}{E_{100/n^2}^{\infty}_{150}}$$

$$= \frac{-\frac{100}{200} - (-\frac{100}{150})}{-\frac{100}{\infty} - (-\frac{100}{150})} = \frac{1}{6} \times \frac{3}{2}$$

$$= 1/4$$

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iv) Prob that one tube lasts more than 150 hrs is 2/3.

$$P(\text{all } n \text{ tubes last}) = (2/3)^n$$

we want prob = 0.5

$$(2/3)^n = 0.5$$

$$n \ln(2/3) = \ln(0.5)$$

$$n = \frac{\ln(0.5)}{\ln(2/3)} = 1.71$$

n must be whole no. $\therefore$ max no. of tubes is n=1

$$Q13) P(|n-4| > 40) \leq 1/4?$$

$p(n) = 2^n$
 n th trial p^p $p(1-p)^{n-1}$ typical of geometric distribution

$$P(|n-3| \leq 2) = P(0 \leq n \leq 4)$$

$$u = 1/p, \quad \sigma^2 = 1-p/p^2$$

$$2 = 1/p \Rightarrow p = 1/2$$

$$\text{Var}(n) = 1 - 1/2 / (1/2)^2 = \frac{0.5}{0.25} = 2$$

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Q. For $p = 1/2$

$$L \geq \frac{2}{6} \Rightarrow \frac{2}{12} = \frac{1}{6}$$

$$P(1 \leq L \leq 2) = 1 - \frac{1}{(1/2)^2} = 1 - 1/2 = 1/2$$

Asked Prob

$$P(0 \leq n \leq 4) = P(n=1) + P(n=2) + P(n=3) + P(n=4)$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{15}{16}$$

$$= p + (1-p)p + (1-p)^2 p + (1-p)^3 p$$

$$= 1/2 + 1/4 + 1/8 + 1/16 = 15/16$$

(i) Each time man selects a key, the prob of selecting the correct key is $1/n$

The prob of failing to select the correct key is $n-1/n$

$$(ii) E(n) = 1/p = 1/(1/n) = n$$

$$Var(n) = 1 - p/p^2 = \frac{1 - 1/n}{(1/n)^2} = \frac{n-1}{1/n} = n(n-1)$$

(iii) If the man eliminates the wrong key of he picks n keys then $n-1$ so no until correct key is selected

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$$E = \frac{1+2+\dots+n}{n} = \frac{n+1}{2}$$

$$Var = \frac{(n^2-1)}{12}$$

(15)

$$y = x^2 + 2x$$

x	1	2	3	4
$P(x)$	0.1	0.2	0.5	0.1
$y = x^2 + 2x$	3	8	15	24

$$f_y = P(y=y) = P(x)$$

y	$P(y)$	$CDF [F_y = P(Y \leq y)]$
3	0.1	0.1
8	0.2	$0.1 + 0.2 = 0.3$
15	0.5	$0.3 + 0.5 = 0.8$
24	0.1	$0.8 + 0.1 = 0.9$

$$f_y = \begin{cases} 0.1, & y=3 \\ 0.3, & 3 \leq y \leq 8 \\ 0.8, & 8 \leq y \leq 15 \\ 1, & 15 \leq y \leq 24 \end{cases}$$

$$E(y) = \sum y \cdot P(y) = 3 \times 0.1 + 8 \times 0.2 + 15 \times 0.5 + 24 \times 0.1 = 11.8$$

$$\begin{aligned} Var(y) &= E(y^2) - [E(y)]^2 \\ &= 9 \times 0.1 + 64 \times 0.2 + 225 \times 0.5 + 576 \times 0.1 - (11.8)^2 \\ &= 44.56 \end{aligned}$$